We have now spent a lot of time and effort answering the question:

"Given a function f, how do we find f'?"

We now turn this question around:

"What does the existence of f' tell us about f?"

As we will soon see, it turns out that even a little knowledge of f' tells us *a lot* about f, but extracting information about f from f' does not look so simple. For example, consider the following theorem:

Theorem. If f is constant on an interval I, then $f'(x) = 0$ for all x in I.

Is the converse of this theorem true? That is, if $f'(x) = 0$ for all x in an interval I, must f be constant on I?

Physically, this seems obvious - if the velocity of a particle is always 0, then surely the particle must be at rest. However, this seems really hard to prove: the hypothesis $f'(x) = 0$ only means that

$$\lim_{h\to 0} \frac{f(x+h) - f(x)}{h} = 0,$$

and it is not at all obvious how one can use the information about the limit to derive information about the function.

The proof of the converse to the theorem above, as well as many other facts of the same sort, can all be derived from the *Mean Value Theorem*, which is one of the most important theorems in calculus:

Theorem. (Mean Value Theorem) If a function f is

(1) continuous on $[a,b]$, and

(2) differentiable on (a,b),

then there is a number $c \in (a,b)$ such that

$$f'(c) = \frac{f(b)-f(a)}{b-a}.$$

That is, there is some $c \in (a,b)$ such that the instantaneous rate of change of f at c is equal to the average or "mean" rate of change of f over $[a,b]$.

Example. If you travel 60 miles in one hour, then at some time you must have been traveling exactly 60 miles per hour.

Geometrically, the Mean Value Theorem says that some tangent line is parallel to the secant line between $(a,f(a))$ and $(b,f(b))$.

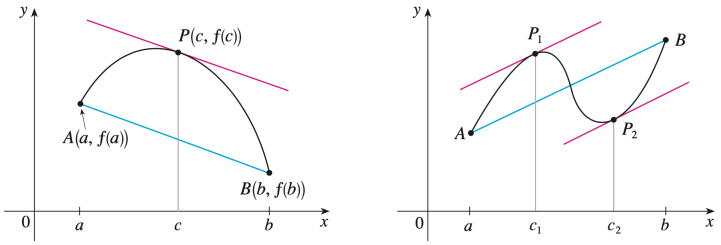


Figure 1: Illustration of the Mean Value Theorem.

Note that continuity at the endpoints is required, as is shown in the example below.

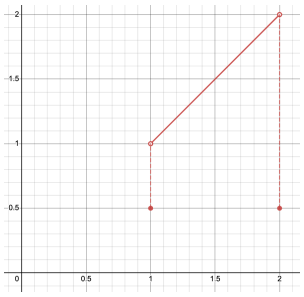


Figure 2: The function above is differentiable on $(1,2)$, but discontinuous at the endpoints. The slope of the secant line connecting $(1,0.5)$ to $(2,0.5)$ is zero, while the slope of the tangent line at every point in the interval $(1,2)$ is 1.

1. Suppose you travel 38 miles in two hours. According to the Mean Value Theorem, at some point you must have been traveling at what speed?

2. Consider the function $f(x) = 2x^2 - x$ on the interval $[1,4]$. What does the Mean Value Theorem tell you about this function?

3. Here is an interesting application of the Mean Value Theorem: The speed limit on a certain road is 45 mph. The road passes through a $\frac{1}{4}$-mile long tunnel. A radar device measures the car's speed to be 42 mph as it enters the tunnel and and another device measures the car's speed to be 40 mph as it exits 15 seconds later. Despite both measurements being under the speed limit, the driver is issued a speeding ticket. Why?

Proof of the Mean Value Theorem

To prove the Mean Value Theorem, we begin with a special case.

Theorem. (Rolle's Theorem) If f is

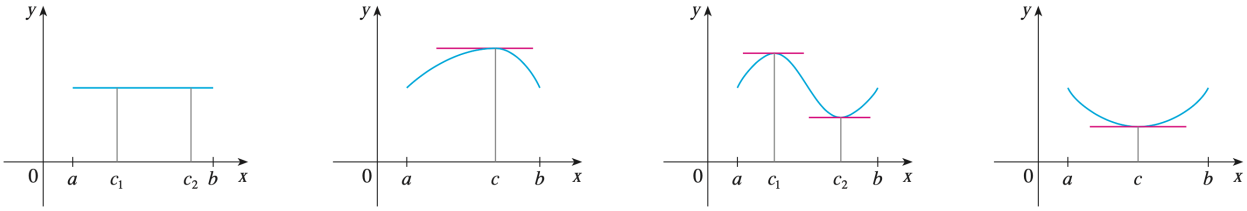
(1) continuous on $[a,b]$,

(2) differentiable on (a,b), and

(3) if $f(a) = f(b)$,

then there is a number $c \in (a,b)$ such that $f'(c) = 0$.

By Hypothesis 3 above, the secant line between $(a, f(a))$ and $(b, f(b))$ is parallel to the x axis, so in this special case the Mean Value Theorem reduces to the statement that there is a point c in (a,b) at which the tangent line at $(c, f(c))$ is also parallel to the x axis. Several examples are shown below.



(a) (b) (c) (d)

Example. You throw a ball directly upward and catch it. What does Rolle's Theorem tell you about this situation?

Justify each statement below to complete the proof of Rolle's Theorem.

Proof.

1. f has both a maximum and a minimum on $[a,b]$.

2. If either the maximum or the minimum occurs in (a,b), then Rolle's Theorem is true.

3. If the maximum and minimum both occur at the endpoints, then Rolle's Theorem is true. *Hint: What must be true about a function with $f(a) = f(b)$ if its maximum and minimum both occur at the end points?*

   This shows that Rolle's theorem is true in all cases, completing the proof. □

4. Is Rolle's Theorem still true if we relax the hypothesis that f is differentiable at *every* point of (a,b)? Explain why or why not. *Hint: Consider the function $f(x) = |x|$ on $[-1,1]$.*

◇ **Check your work with your instructor before moving on.**

Proof of the Mean Value Theorem

We now prove the Mean Value Theorem using Rolle's Theorem.

Proof. To prove the Mean Value Theorem using Rolle's Theorem, it is sufficient to produce a function h which satisfies the hypotheses of Rolle's Theorem, and for which $h'(x) = f'(x) - \frac{f(b)-f(a)}{b-a}$, so that the conclusion $h'(x) = 0$ of Rolle's Theorem yields $f'(x) = \frac{f(b)-f(a)}{b-a}$.

1. Find a function g such that $g'(x) = \frac{f(b)-f(a)}{b-a}$ for all x. *Hint: Note that* $\frac{f(b)-f(a)}{b-a}$ *is just a real number. If you can find a function g such that* $g'(x) = 2$ *for all x, then you can find a function g such that* $g'(x) = \frac{f(b)-f(a)}{b-a}$ *for all x.*

2. Show that $h = f - g$ satisfies the hypotheses of Rolle's Theorem.

3. Show that $h'(x) = f'(x) - \frac{f(b)-f(a)}{b-a}$ and apply Rolle's Theorem to finish the proof.

♢ **Check your work with your instructor.**